

A spatially explicit dataset of production, trade, and corporate control in the Brazilian soy sector

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Abstract

TODO: abstract.

Keywords: soy, land footprint, telecoupling, MRIO, FABIO, Brazil, deforestation

1 Background & Summary

- Relevance of the topic of understanding dynamics in the Brazilian soy sector from both an environmental and a socio-economic view
- Review of existing data sets
- Identification of data gaps
- Justification for the data set presented in this work, its relevance and potential applications

2 Methods

The methods section is too long as it is now, in particular after the sub-section 'Method overview'. We will likely need to move some parts into an Appendix. In particular, as also Siwar's company component is entirely missing. Still to be decided, how this will be implemented.

This section describes ...

2.1 Data sources

The model is built entirely from publicly and openly available data. Four groups of inputs are combined: (i) Brazilian subnational statistics on soy production, processing, domestic use and trade; (ii) national commodity balances and feed parameters; (iii) administrative and environmental geographies; and (iv) the global input-output backends and the independent validation reference series. Unless otherwise stated, the

series are annual, cover the period 2001–2022 and are broken down by Brazil’s approximately 5,570 municipalities (*municípios*). Where a source exists for only a single reference year, that year is stated and the layer is held constant. Table 1 summarises every input, and all access links are collected in Appendix I. A downloader for the machine-retrievable sources is included with the code.

All input data are publicly and freely available from the providers listed in Appendix I. They are not redistributed with the code because of differing licences; a download script covers the machine-retrievable sources, and the datasets requiring manual retrieval (ABIOVE, ANP, POF, MapBiomass tiles) are documented step by step in the repository. The intermediate and final model outputs underlying the figures will be deposited on Zenodo upon publication.

2.1.1 Brazilian subnational statistics

Production and population (IBGE): Municipal soybean production and the planted and harvested area data come from the SIDRA system (Table 1612), obtained via its public API. The resident population (Table 6579; 2022 Census) serves as a proxy for household soy oil consumption.

Livestock (IBGE): Municipal herds for ten animal categories (Table 3939) and milked cow counts (Table 94) provide the feed demand base. Feedlot cattle, last reported in the 2006 agricultural census (the 2017 census dropped the confinement question), are carried forward from the 2006 municipal pattern using the national ABIEC/Athenagro confinement series (see Methods).

Processing and biodiesel capacity (ABIOVE, ANP): Soy crushing and refining capacity comes from two manually downloaded ABIOVE products: the consolidated state-level installed capacities and the annual plant roster surveys (*Pesquisa de Capacidade*), whose changing sheet layouts are handled by a per-year dispatcher in the code. Refining and bottling capacities come from the refining sheets of the same products. Years for which surveys are missing from the ABIOVE archive (2000–2002, 2016–2017 and 2021) are replaced with the previous available survey. Biodiesel plant capacity and the regional soy feedstock share are taken from the ANP Statistical Yearbook.

Household consumption and storage (IBGE): State-level per capita soy oil acquisition from the Household Budget Survey (POF 2017–18, the most recent) is used to distribute national food oil use and is held constant across years. Grain storage capacity, which allocates stock changes, is aggregated from the georeferenced warehouse registry underlying the IBGE Stocks Survey (2014 snapshot, held constant).

Municipal trade (COMEX/MDIC): Municipality-level exports and imports are sourced from the COMEX Stat customs database and filtered by the HS4 soy codes 1201 (beans), 1507 (oil) and 2304 (cake).

2.1.2 National balances and feed parameters (FAO)

Commodity balances: The national supply-use balances against which all municipal totals are reconciled are the FAO Food Balance Sheets for Brazil, items 2555 (soybeans), 2571 (soybean oil) and 2590 (soybean cake), taken from the FAOSTAT bulk download.

Bilateral trade: Re-export disentangling draws on the FAOSTAT Detailed Trade Matrix (items 236, 237 and 238 for beans, oil and cake).

Livestock systems and feed rates: Herds are categorised by production system using the Gridded Livestock of the World rasters (GLW3, reference year 2010) [1] and the GLEAM ruminant system map. They are then converted to soy feed demand using GLEAM per-animal intake rates [2]. These are static layers: the resulting system shares are held constant while herd sizes remain annual (see Methods).

2.1.3 Geographies

Boundaries: Municipal polygons are IBGE administrative meshes retrieved with the `geobr` package [3]; state outlines for mapping use GADM v3.6.

Biomes and frontier: The footprint origin is attributed to biomes using the IBGE 1:250,000 biome layer. The agricultural frontier is the Amazon biome plus MATOPIBA, operationalised as the Cerrado biome within Maranhão, Tocantins, Piauí and Bahia, closely matching the official Embrapa delineation [4].

Grid refinement (optional): A 30 m visual refinement uses MapBiomass annual soy tiles [5], exported via Google Earth Engine. This refinement is illustrative only and does not affect the load-bearing capacity of the results.

Transport network: The reported model routes flows based on straight-line distances between municipal centres (see Methods). **Four multimodal layers — OpenStreetMap roads, DNIT waterways, ANTT rail lines and cargo, and ANTAQ ports and port cargo — support an experimental cost-based transport variant stored in the repository and used for descriptive analyses. These layers do not affect the reported results.**

2.1.4 Global models and reference series

FABIO. Subnational flows are incorporated into the Food and Agriculture Biomass Input–Output model [6]. From 2010 onwards the version 2 build is used (2010–2023; 22,263 processes, 181 regions $\times$ 123 items, with 123 environmental extensions). As v2 does not cover years before 2010, the period 2001–2009 uses the corresponding inputs of the v1.1 build: bilateral trade details, land use extensions, feedstock optimisation results and v1 balanced trade and commodity balance sheets. Version bridging and the quantified v1.1/v2 seam are described in the 'Methods' section.

EXIOBASE. The non-soy background economy is EXIOBASE 3 [7] in the product-by-product transformation (2000–2022; 49 regions $\times$ 200 products), hybridised with FABIO for the non-food component of the soy footprint.

Trase (validation only). Modelled origin-to-importer flows are validated against the Trase Brazil soy composite v2.6.1 (2004–2022), which links producing municipalities to importing countries from per-shipment customs and logistics records [8, 9]. No Trase data enter the model itself.

Table 1 Input datasets, by role. All are publicly and freely available; access links are collected in Appendix I. “Munis” = Brazilian municipalities; “exp.” = experimental variant only.

Dataset	Provider	Role	Coverage
Soy production, area (t.1612)	IBGE (SIDRA)	production	munis, annual
Population (t.6579/9514)	IBGE (SIDRA)	food-use proxy	munis, annual
Herds (t.3939), milk cows (t.94)	IBGE (SIDRA)	feed demand	munis, annual
Feedlot cattle (t.919)	IBGE (Census)	feed demand	munis, 2006 pattern
Crush/refining capacity	ABIOVE	processing	munis, annual
Biodiesel capacity	ANP	other (biodiesel) use	munis, annual
Soy-oil acquisition (POF)	IBGE	food-oil allocation	states, 2017–18
Grain storage (warehouse registry)	IBGE	stock allocation	munis, 2014 static
Commodity balance (FBS)	FAO (FAO-STAT)	national reconciliation	Brazil, annual
Gridded livestock (GLW3)	FAO (Data-verse)	system split	10 km, 2010
Production systems, feed (GLEAM)	FAO	feed rates	10 km, static
Municipal trade	COMEX/MDIC	exports/imports	munis, annual
Road network	OSM (Geofabrik)	transport (exp.)	national
Waterways	DNIT	transport (exp.)	national
Rail lines, stations, cargo	ANTT	transport (exp.)	national, 2006–21
Ports, port cargo	ANTAQ	transport (exp.)	national, 2017–22
Municipal boundaries	IBGE (geobr)	geometry	munis
State boundaries	GADM v3.6	mapping	states
Biomes 1:250k	IBGE	origin attribution	national
Soy land tiles	MapBiomass (GEE)	grid refinement	30 m, annual
FABIO v2 (MRIO)	WU Vienna	supply-chain nesting, 2010–2022	global, 2010–23
FABIO v1.1 (MRIO)	WU Vienna	supply-chain nesting, 2001–2010	global, 1986–2013
EXIOBASE 3 pxp	EXIOBASE	tracing soy flows to non-food uses	global, 2000–22
Detailed trade matrix	FAO (FAO-STAT)	re-exports/trade	Brazil, annual
Trase composite v2.6.1	Trase	validation reference series	munis, 2004–22

3 Methods

Our approach couples a subnational (municipality-level) reconstruction of Brazil’s soy supply chain with the global multi-regional input–output (MRIO) frameworks FABIO [6] and EXIOBASE [7]. The model allows calculating the material and land-use footprints of soy products consumed in countries worldwide and tracing them back to Brazilian municipalities of origin. This is realized on an annual basis for the period 2001–2022. The subnational tracing framework generalizes the single-year (2013) open model of Trsek [10] to an annual series.

The method is implemented in six stages, summarized in Figure 1: (i) municipal soy accounting; (ii) livestock-system disaggregation and feed demand; (iii) reconciliation with national commodity balances; (iv) allocation of physical flows between municipalities by cost-minimizing optimized transport; (v) linkage of origin municipalities to consuming countries, with re-export disentangling; and (vi) nesting into FABIO and land-use footprint accounting. The results are validated against the independent Trase supply-chain reference series [8] and a simple downscaling baseline.

The subsections 3.1–3.6 below correspond to stages 1–6 and describe the approach step-by-step, including input data, calculation procedures and outcomes. All equations, data-processing choices and implementation details are presented in Appendices A–H; a script-level mapping of each stage to the released pipeline code is provided with the GitHub repository (reference/URL).

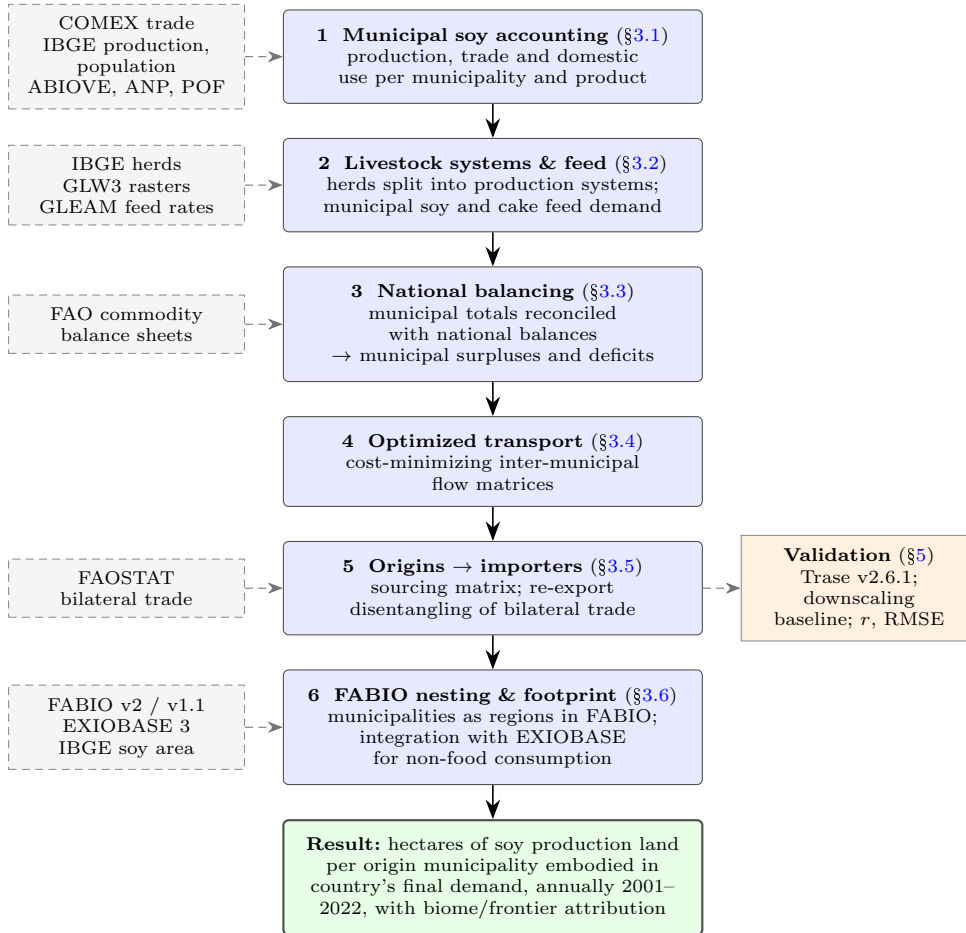


Fig. 1 Workflow of the study. Solid boxes are the six model stages (Sections 3.1–3.6); dashed boxes are the main data inputs at each stage; the orange box is the independent validation of the reconstructed flows (Section 5). Stages 1–5 reconstruct Brazil’s subnational soy supply chain; stage 6 embeds it in the global FABIO/EXIOBASE input–output framework and computes the land-use footprint.

3.1 Municipal soy accounting

In the first stage, a complete account of supply, trade and domestic use is compiled for all Brazilian municipalities for each of the three soy products: soybeans, soy oil and soy cake (the protein meal remaining after the beans are crushed for oil). Exports and imports are obtained from COMEX customs records at municipality level, production data comes from IBGE municipal surveys, and processing. We take storage and biodiesel infrastructure data from ABIOVE, georeferenced warehouse registers and ANP. Each item of the national FAO commodity balance for domestic use (crush, food, seed, other uses (e.g. biodiesel) and stock changes) is then distributed across municipalities according to the weight most closely tied to that use (see Table 2): installed crushing capacity for the crush, household soy oil consumption for food use, and so on. The outcome is a municipal supply–use account for beans, oil and cake that reproduces the national commodity balance. The data cleaning process, derivation of each weight, and treatment of stock changes are detailed in Appendix A.

Table 2 Domestic-use allocation: each item is a national commodity-balance total distributed across municipalities in proportion to the stated weight w_m (Appendix A).

Use item	Allocation weight w_m
Processing (crush)	installed crushing capacity
Food (beans, oil)	per-capita soy-oil acquisition $\times$ population
Other (biodiesel)	soy-attributable biodiesel capacity
Seed	soybean production
Stock change	storage capacity (beans), production (oil, cake); addition $\rightarrow$ use, withdrawal $\rightarrow$ supply

3.2 Livestock systems and feed demand

The next step is to estimate the municipal feed demand for soybeans and soy cake. Since the amount of soy consumed by each animal varies greatly depending on the livestock production system, the IBGE’s herd counts at municipal level are first divided into production systems (extensive, intensive, and industrial monogastrics; grassland- and mixed-based ruminants; dairy and meat) using the 2010 FAO gridded livestock maps (GLW3 and GLEAM). The number of animals in each system is then multiplied by the amount of soy and soy cake consumed by each animal in that system, as determined by GLEAM. The bottom-up totals are then rescaled to the national feed use reported by the FAO (see Figure 2). The outcome is the municipal feed demand for beans and cake by species and production system, which is consistent with the national balance. The splitting formulas and a discussion of the implications of keeping the 2010 system shares fixed throughout the study period are provided in Appendix B.

3.3 National balancing

Next, the municipal accounts are reconciled with the national supply–use identity of each product (production + imports + stock withdrawal = exports + food + feed + seed + processing + other). Municipal trade is harmonised with the FABIO/FAOSTAT bilateral trade classification. Any remaining national residual is absorbed into the stock term and each balance item is proportionally rescaled so that every municipal aggregate matches its national total exactly. The outcome is a balanced set of municipal net positions for each product (a surplus where local supply exceeds use, a deficit where use exceeds supply). The supply–use identity, the surplus/deficit decomposition, and the treatment of net stock withdrawals are explained in Appendix C.

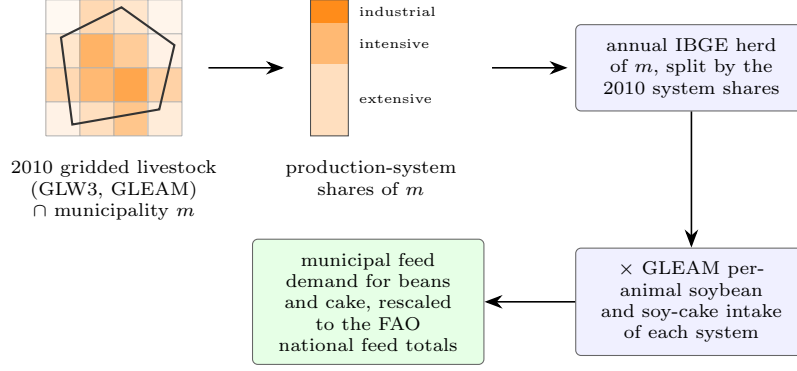


Fig. 2 From herds to municipal soy feed demand. Municipal herd counts (annual) are divided into production systems using the shares implied by the 2010 gridded livestock rasters within each municipality. Each system’s herd is multiplied by its per-animal soy and soy-cake intake, and the totals are then scaled to match the national feed use reported by the FAO.

3.4 Subnational transport allocation

In step 4, municipal surpluses and deficits are connected via optimized transport that minimizes costs. For each product, the inter-municipal flow matrix ($\mathbf{F}^p$) reflects the cheapest way to assign surpluses to deficits while conserving the mass balance. The premise is that bulk-commodity logistics minimize freight costs, which increase with distance. Three allocation variants are analyzed in order of increasing supply-chain structure: a pure downscaling baseline, which ignores the domestic network and splits each destination’s imports across municipalities according to their production shares; the Euclidean model, which solves the transportation problem using straight-line distances between municipal centres; and a multimodal variant, which uses Brazil’s actual road, rail and waterway networks for the same flows, taking into account mode-specific freight costs and corridor capacities (all three variants are specified in Appendix D). When compared with the independent Trase reference series, the three models produce similar results at the aggregate level: the pooled municipality×destination correlation averages 0.71, 0.69 and 0.69 respectively over the period from 2010 to 2022, but differences emerge at the level of individual destinations (see Section 5). The reported data use the Euclidean model, as the multimodal variant does not reproduce the reference series better than the Euclidean model in any year, and requires network inputs that are unavailable for parts of the study period. The downscaling baseline produces no inter-municipal flows, which are required for the tracing of processing, feed and domestic consumption in step 5. The outcome is a deterministic inter-municipal flow matrix per soy product whose margins reproduce the municipal surpluses and deficits by construction. The formal problem statement, the solver, and the multimodal results are described in Appendix D.

3.5 From origins to consumers

The COMEX customs records of stage 1 attach two labels to every exported tonne: the municipality that shipped it and the country that first imported it. However, neither label contains the information that the footprint needs: the municipality that grew the soy and the country that finally consumes it. Step 5 tackles both issues. Within Brazil, the transport flows of step 4 serve as the tracing device. Completed with each municipality’s retained supply, they yield sourcing shares that trace every exported (and domestically consumed) tonne back to its origin municipality (Figure 3). Oil and cake are traced one step further back to the municipalities where the beans were grown. Moving forwards across borders, exported tonnes often pass through intermediate countries before being consumed. The FAOSTAT bilateral trade data are therefore corrected for re-exports [11]: **each tonne is followed through the trade network until it reaches the country where it is finally consumed** (Figure 3). In this calculation, Brazil is represented not as a single country, both with municipality-level

resolution. The outcome is an origin-to-consumption matrix linking every producing municipality to every consuming country. The equations, the underlying proportional-mixing assumption and its numerical implementation are given in Appendix E.

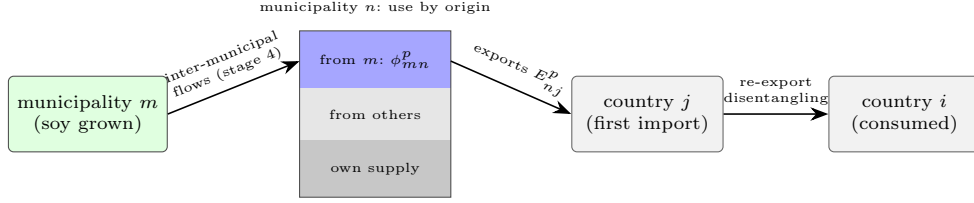


Fig. 3 Tracing a tonne from origin to final consumer. The soy grown in municipality m reaches municipality n through the inter-municipal flows of step 4, so the share ϕ_{mn}^p of n 's use is attributed to m (highlighted). The observed exports of n carry that share to the country of first import j , and the re-export disentangling reattributes it to the country of final consumption i . The corresponding equations are given in Appendix E.

3.6 Nesting into FABIO and footprint accounting

In the final stage, the reconstructed municipal soy economy is embedded in the FABIO physical MRIO, in which a process is a region $\times$ commodity unit. Brazil's three national soy commodities and processes are removed and replaced by their $\sim 16,700$ municipal counterparts, so that municipalities enter the global model as additional regions, each carrying its own production, soy area, feed use and origin-resolved trade from the previous steps (Figure 4). The regional accounts are then linked through the re-export-corrected trade of step 5 into a single global input-output model, following the published FABIO procedure [6]. Soy used outside the food system, for example as biodiesel, is traced onwards by coupling FABIO to the EXIOBASE model covering non-food supply chains in the wider economy [7]. From 2010 onwards the model runs on the FABIO v2 backend; for 2001–2009 it switches to inputs from the FABIO v1.1 build, and the difference between the two versions is quantified on the overlap years 2010–2013. The assembly of the nested MRIO, the numerical safeguards, and the backend bridging and difference analysis are documented in Appendix F.

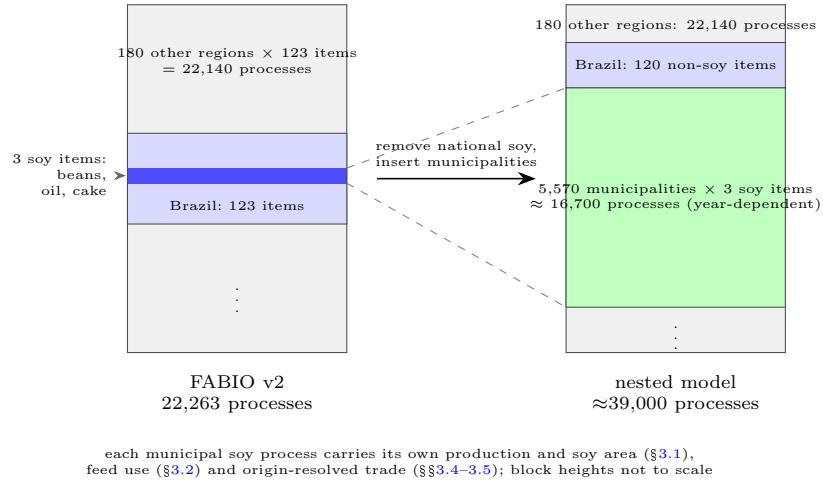


Fig. 4 Nesting municipal soy into the global input-output system: the three national Brazilian soy commodities are removed from the process list and replaced by one soy process set per municipality, populated with the municipal accounts of steps 1–5; all other regions, and Brazil's remaining commodities, are untouched. The structure of the combined model is detailed in Appendix F.

Land-use footprint accounting

The direct land use of each process is its harvested soy area, taken from IBGE for the municipal soy processes and from the FABIO land-use extension for all others. Dividing by output gives a land intensity per unit of product, which the footprint calculation traces through the supply chains to each country’s final demand, both for food and non-food uses. Each municipality is assigned to a biome (IBGE 1:250,000 layer), and the agricultural frontier is defined as the Amazon biome plus MATOPIBA. The result is hectares of soy land in each origin municipality embodied in each country’s final demand, per year, with biome and frontier attribution. The treatment of stocks, footprints by consumed product, and the optional 30 m grid refinement are described in Appendix [G](#).

4 Data records

ADD: data description.

SOYPRINT database (soyprint.duckdb). Fact tables cover 2000–2022 and are hive-partitioned by year in the Parquet layer.

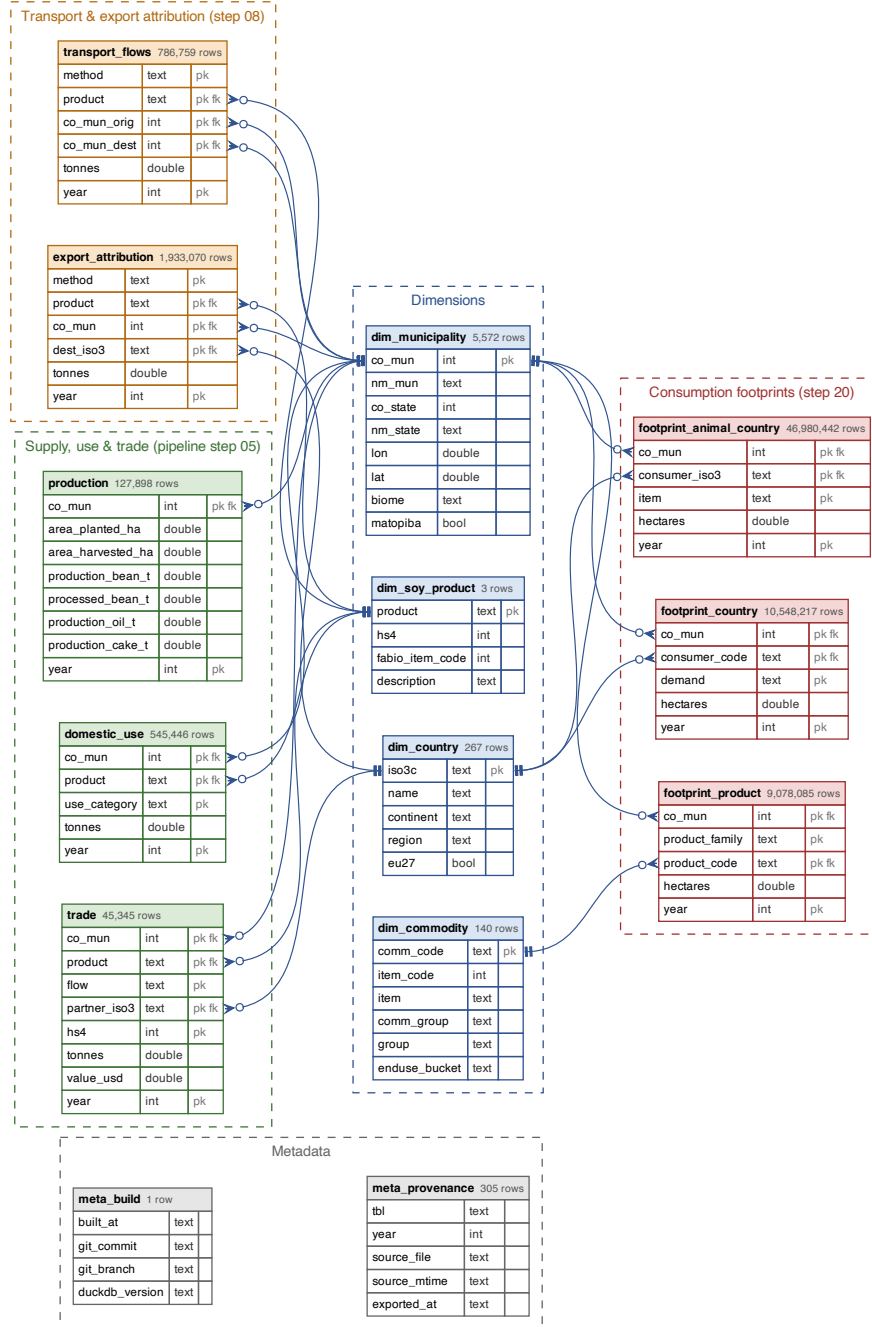


Fig. 5 Schema of the released results database: eight fact tables, grouped by the pipeline stage that produces them, linked to four dimension tables, with build and provenance metadata alongside. Each field is listed with its type, and primary and foreign keys are marked (pk, fk). Fact tables are hive-partitioned by year in the Parquet layer.

4.1 Headline results

Four outputs illustrate what the pipeline delivers. The footprint grew from 9.4 Mha in 2001 to 41.0 Mha in 2022, with its origin shifting from the southern states to the

Cerrado frontier (see Fig. 8). During the same period, the EU-27 and China exchanged roles as the dominant consumer, with this change occurring in 2009–2010 (Fig. 6). Most of the footprint reaches consumers in commodity form. From 2010 onwards, around a quarter is embodied in animal products (Fig. 7). At the finest level, the footprint can be displayed as probability maps per destination on 30 m MapBiomass soy pixels (Fig. 9).

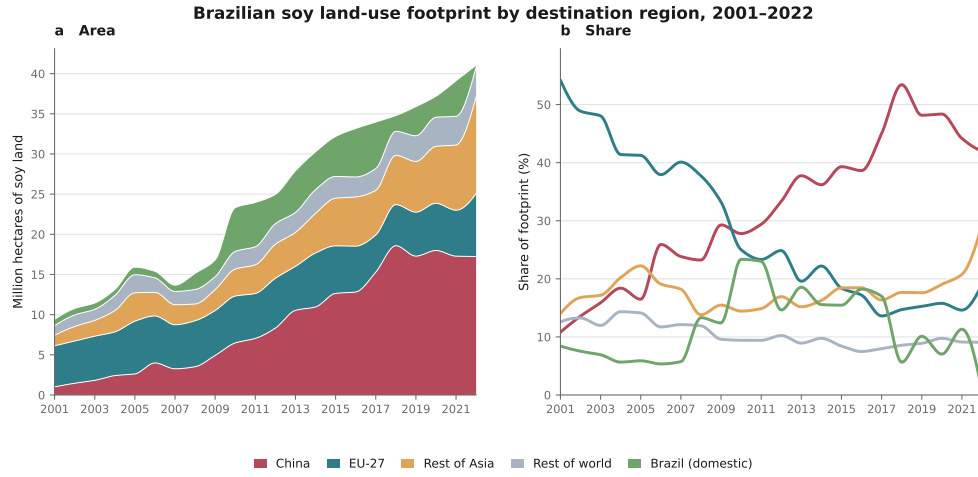


Fig. 6 Footprint by destination region, 2001–2022: (a) absolute footprint, (b) regional shares.

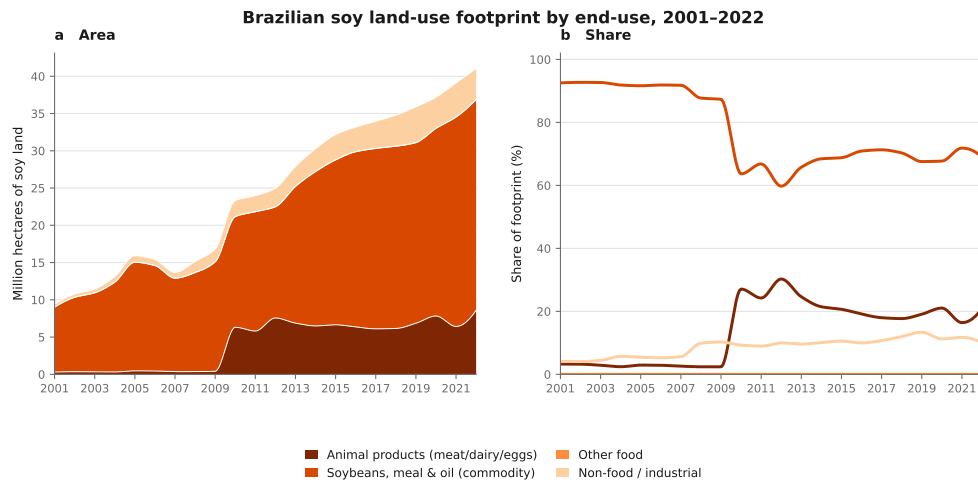


Fig. 7 Footprint by end-use, 2001–2022: (a) absolute footprint, (b) end-use shares.

Brazilian soy land-use footprint by destination, municipal origin, 2001 vs 2022 Hectares of soy land embodied in consumption (F_{mass}). All panels share one colour scale; sqrt-scaled.

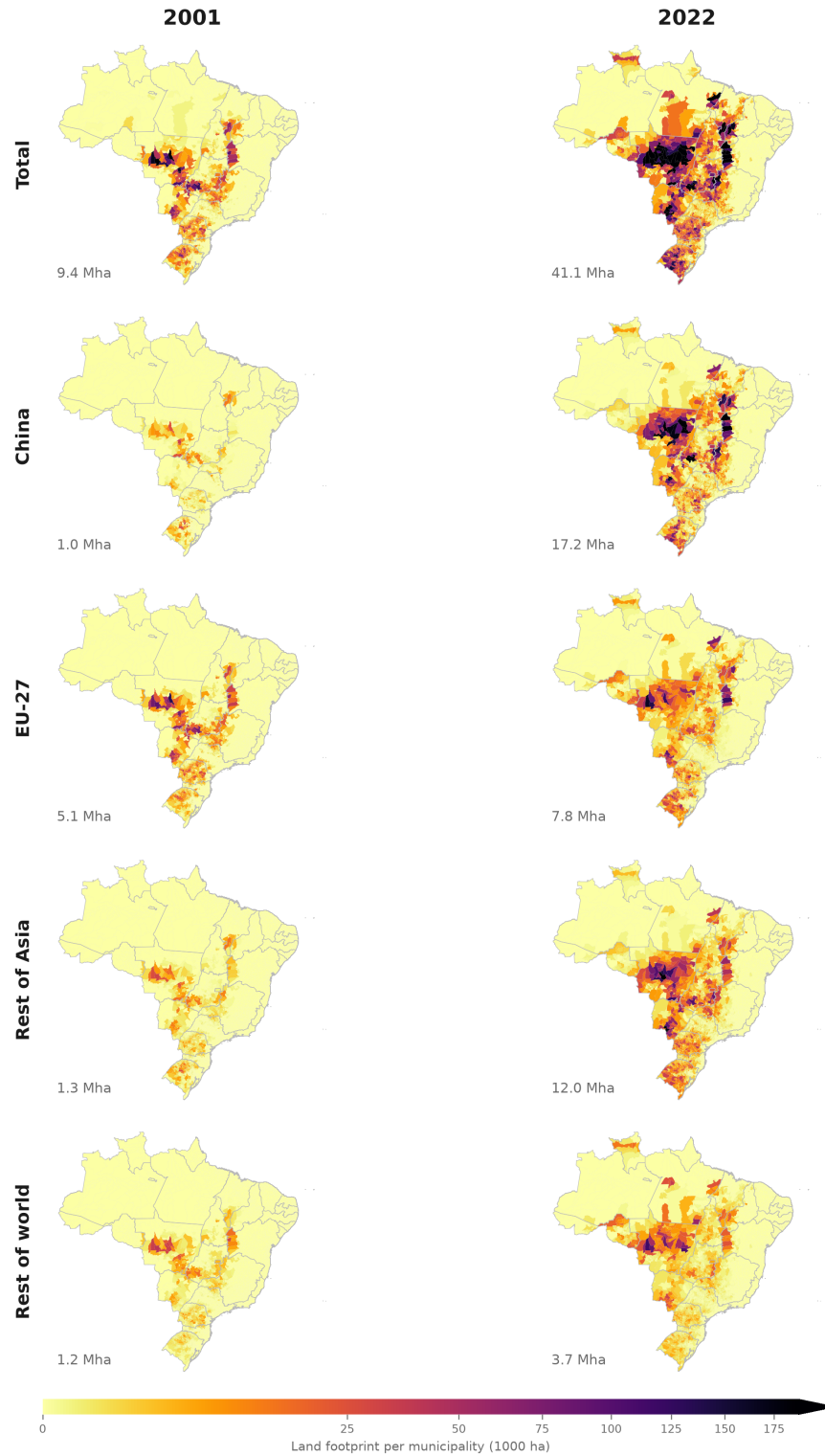


Fig. 8 Municipal origin of the Brazilian soy land-use footprint, 2001 versus 2022, for the total and the four major destination groups; all panels share one square-root colour scale. An interactive year-by-year version is available at <https://lizaburiak.github.io/soyprint-footprint-map/>.

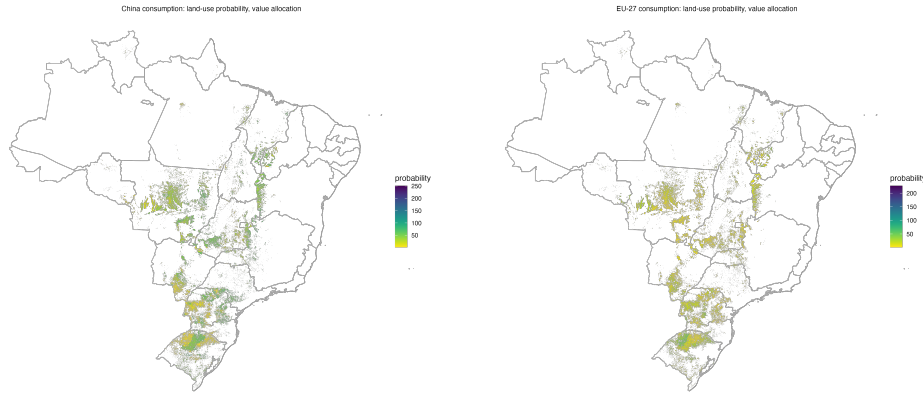


Fig. 9 Probability maps of the 2022 footprint on 30 m MapBiomass soy pixels, for China (left) and the EU-27 (right); colour gives the pixel's probability of serving the destination's consumption (value allocation).

5 Technical validation

The technical validation focuses on two aspects. First, a comparison of results against the method from the 'Trase' initiative as an established approach that produces municipal-level soy production, trade and footprint accounts for Brazil (Section 5.1). Second, an assessment of the sensitivity of our results against changes in input assumptions (Section 5.2).

5.1 Comparison with results from 'Trase'

This validation step addresses the subnational part of the calculation pipeline and the allocation to the first importing countries. The amount of soy and corresponding land use that further propagates through global supply chains to final consumers is documented in FABIO, a peer-reviewed global physical input-output database, which has its own documentation of construction principles and validation [6].

The newly constructed origin-to-importer flows are evaluated against the independent Trase Brazil-soy dataset (v2.6.1), which links municipalities of production to countries of first import using per-shipment customs and logistics records [8, 9]. Trase's modeling approach is similar to the one employed here, integrating a wide range of subnational data on production, usage, logistics and trade. Its key differentiator is the usage of per-shipment trade data (customs records, bills of landing), which is linked to individual trading companies and their geo-localized export hubs (e.g., farms, storage, processing or wholesale facilities) via tax registration data. This data is proprietary and thus not available for an open source pipeline as presented here.

Agreement is measured by the Pearson correlation between the modelled and the reference series flows, computed over all matched municipality-destination pairs of a year (pooled), for the pairs of one importing country at a time (per-destination), or on $\log_{10}$ flows in the rocket plots presented below. Two features organise the comparison. First, agreement depends on flow size: the model and the reference series agree closely on the large flows that carry the main part of tonnage and less on the small ones, which are numerous but negligible in volume. Second, from 2019 onward the reference series itself changes: Trase switched its per-shipment source, and part of the post-2018 disagreement is a property of the reference series rather than of the model.

The dependence on volume shapes the choice of the headline metric. Nearly half of the matched flows (46%) are below 1,000 t, and together they carry 1.3% of the volume (Figure 11); a plain average over destinations therefore lets the noise of negligible flows dominate, giving e.g. Cuba the same weight as China. We therefore use the headline metric of volume-weighted per-destination correlation: for each importing country the modelled map of origin municipalities is correlated with the map Trase reports for

that importer, and the per-importer scores are averaged with the importer’s share of the year’s volume as weight (Figure 10).

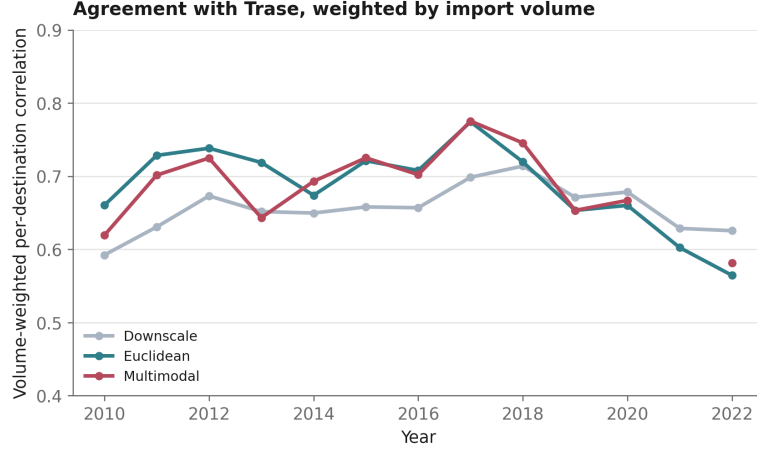


Fig. 10 Volume-weighted per-destination correlation with the Trase reference series, 2010–2022, for the three allocation variants (the multimodal variant has no 2021 solve). Weights are each importer’s share of the year’s Trase-recorded import volume; Brazil (domestic use) is excluded. The supply-chain models exceed the downscaling baseline in every year up to 2018; after the reference series’s 2019 change of source the ordering inverts.

All three allocation variants are compared on this metric: the downscaling baseline, in which every importer draws from the same municipal map and only totals differ, the Euclidean model, and the multimodal transport variant. The weighted correlation of the Euclidean model runs between 0.66 and 0.77 over 2010–2018 and exceeds the baseline in every year of that period, by 0.05 to 0.10 in most years, with the gap closing in 2018; the multimodal variant tracks the Euclidean model within 0.03 in most years and 0.08 at worst (2013). Since import totals are identical in all variants, this gap is exactly the information added by the supply-chain structure: different importers source from different parts of Brazil, and only the supply-chain models reproduce such differences. After 2018 all variants decline and the ordering inverts, which we trace to change of source data (see below).

The average conceals large differences between destinations. China, which takes most of the exported tonnage, scores 0.69 to 0.86 across years and Spain reaches 0.93, while for the smallest importers the scores scatter widely from year to year, from slightly negative to above 0.9, with no persistence. The annex (Section H.2) reports the full per-importer trajectories for the four importers that cover 70% of the volume and for the five smallest importers matched in every year.

Part of the distance that remains to the reference series is a data limitation rather than a model choice. Installed crushing capacity is published by ABIOVE as state totals together with a roster of plant locations, with no capacities for individual plants, so the pipeline has to fix an allocation rule and divides each state’s capacity equally among its plants; Section 5.2 quantifies what this assumption implies. Trase, by contrast, observes individual shipments: per-shipment customs records name the exporting company, and the company’s tax registration identifies its silos and crushing facilities and their municipalities, so most of each flow is anchored to a specific logistics hub before any modelling.

Agreement with Trase generally rises with flow size (Figure 11). Pooled over the 91,873 matched municipality×destination×year flows of 2010–2022, which span six orders of magnitude, the correlation of log flows with Trase is 0.68: large flows converge onto the 1:1 diagonal, small flows scatter. The same gradient shows in the per-destination scores: the mean per-destination correlation rises from 0.29 in the lower half of import volumes to 0.53 in the upper half.

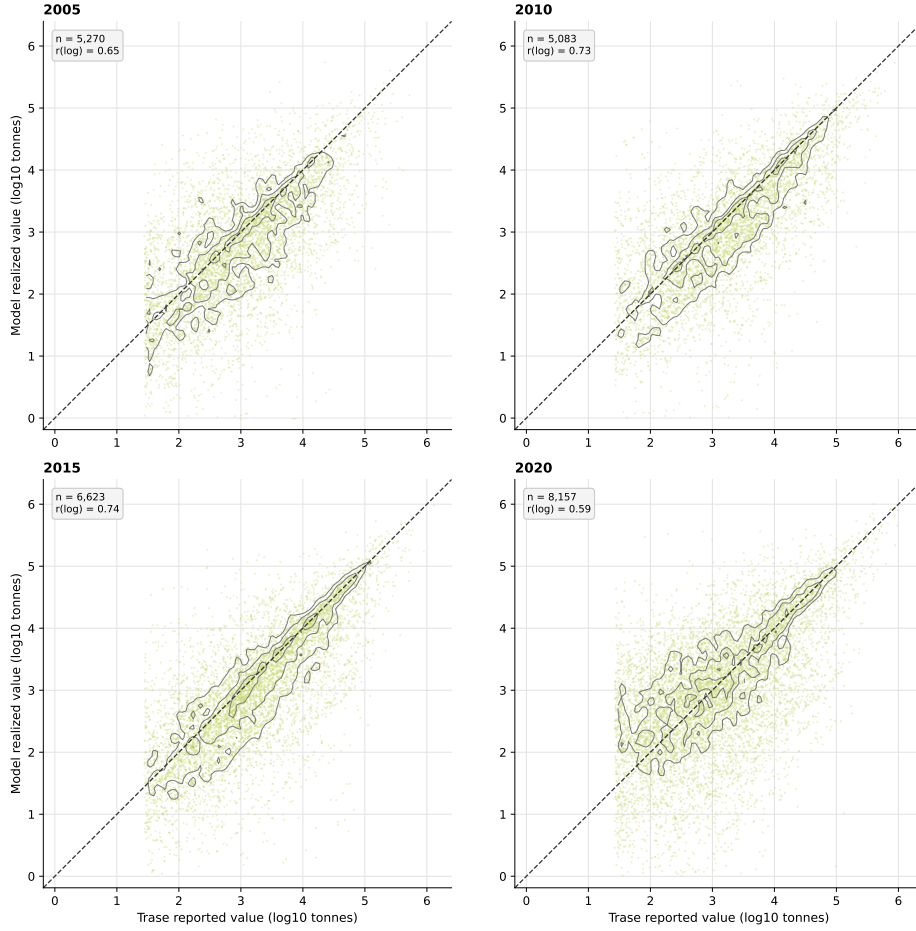


Fig. 11 Agreement with the reference series grows with flow size (“rocket plots”). Every modelled municipality×destination soy export flow is plotted against the corresponding Trase flow (both $\log_{10}$ tonnes; flows below 1 t on either side are excluded) for four sample years. Because thousands of points overlap, grey contour lines outline where the point cloud is densest, in the way elevation lines outline a hill; the dashed line is the 1:1 diagonal on which a perfectly reproduced flow would lie. Each panel reports the number of matched flows and the log-scale Pearson correlation. Large flows converge onto the 1:1 diagonal (the “nose”), while small flows scatter (the “plume”).

The weaker agreement visible in the 2020 panel is not specific to that year but marks the whole 2019–2022 block, and its origin is on the reference series side. From 2019 onward Trase switched its per-shipment source to maritime-only bills of lading that record the country of first import rather than the final destination, and the share of its exports with unknown origin municipality roughly doubles [12]. As a consequence, the log-scale correlation decreases from 0.70–0.74 (2010–2017) to 0.59–0.65 (2019–2022), and the small flows drift above the diagonal: below 1,000 t the model reports on average 1.6 times the reference series value in 2019–2020. The destination pattern identifies the mechanism. China, served by direct shipments, is unaffected (mean log deviation 0.00 in 2020), while Slovenia, whose port of Koper is a transshipment hub for landlocked Europe, is the strongest outlier in the opposite direction: the reference series books to the country of first import the tonnage that the customs-based model books to final destinations, which in turn appear mildly over-reported (Thailand, Italy, Germany). Trase itself recommends against comparing its 2004–2018 and 2019–2022 periods directly, and the post-2018 part of the correlation band should be read with that caveat.

5.2 Sensitivity to input assumptions

The second part of our validation tests the sensitivity of two core assumptions required to fill gaps in input data: (a) the within-state location of crushing: ABIOVE reports capacity only as state totals, and our baseline assumption divides each state’s total equally among its plants; (b) the feed and livestock parametrisation: the GLEAM per-system feed rates and the municipal herd composition taken as constant 2010 values across the entire time series.

Each perturbed variant re-runs the full flow reconstruction up to the origin-to-importer flows for a representative year (2019) and is scored against Trase on the pooled municipality $\times$ destination correlation (Euclidean model, base $r = 0.72$). We ran twenty-four variants in total.

For assumption (a) on the allocation of soy crushing capacities from the state to the municipal level, the capacity map is perturbed in three ways:

1. Production-proportional split: each state’s capacity is redistributed across its plant municipalities in proportion to municipal soybean production (1 run);
2. Random splits: individual plant capacities are unobserved, so every plant is given a positive random weight, drawn from an exponential distribution, and each state’s fixed total is divided among its plants in proportion to those weights; this makes every possible split of a state’s total equally likely, and repeating it with ten different random seeds gives ten alternative capacity maps (10 runs);
3. Allocation sharpness: municipal capacities act as weights that divide the national crush total among municipalities. Under the equal split they are identical within a state but still differ between states and where one municipality hosts several plants; raising every capacity to a power γ varies how strongly processing follows those remaining differences, with $\gamma = 2$ concentrating it in the highest-capacity municipalities and $\gamma = 0.5$ spreading it more evenly (2 runs).

For assumption (b), the parametrisation is perturbed in two ways:

1. Random feed rates: the per-head bean and cake feed rates of every livestock system are multiplied by random factors, drawn independently for each system from a lognormal distribution with $\sigma = 0.3$, so that the factors typically lie between 0.75 and 1.35; repeating this with ten different random seeds gives ten parametrisations in which the GLEAM rates vary by realistic amounts in different directions (10 runs). The factors average to one and national feed totals are rescaled to the FAO commodity balance afterwards, so an error applied uniformly to all systems cancels out; what this perturbation tests is the system mix, since an error there shifts feed demand toward the municipalities whose herds contain the altered systems;
2. Herd composition: the 2010-frozen municipal herd composition is replaced by state-average system shares (1 run).

Table 3 illustrates the results of the sensitivity tests. We find that our assumption on the location of soy crushing capacities explains nearly all of the induced variation: the random within-state splits span correlations of 0.67 to 0.74 (spread 0.073), the production-proportional split scores 0.69, and the allocation-sharpness variants span 0.024. In contrast, the feed variants are immaterial: the ten feed-rate draws move the correlation by less than 0.01 and the herd-composition variant by less than 0.001. Adding specific information on soy crushing capacity at the level of single municipalities is thus the main leverage point to further increase the preciseness of our results.

Table 3 Input-assumption sensitivity of the model–Trase correlation (pooled municipality×destination flows, Euclidean model, 2019; base $r = 0.723$).

Assumption	Perturbation	Runs	r range	Spread
Crush location	production-proportional split	1	0.691	–
Crush location	random within-state splits	10	0.671–0.744	0.073
Crush location	allocation sharpness $\gamma \in \{0.5, 2\}$	2	0.706–0.731	0.024
Feed rates	random per-system factors (lognormal, $\sigma = 0.3$)	10	0.717–0.725	0.007
Herd composition	state-average system shares	1	0.722	–

A Municipal soy accounting

Throughout the appendices, m and n index Brazil’s 5,570 municipalities, $p \in \{\text{bean, oil, cake}\}$ the three soy products, and i, j countries; matrices and vectors are set in bold. One operation recurs throughout the subnational reconstruction: a national total is distributed across municipalities in proportion to an observable weight, as listed in Table 2 of the Methods document. The released pipeline checks each such allocation against its national total at run time and halts on any violation.

A.1 Data assembly and domestic-use estimation

Five rules complete the allocation weights of Table 2 and the assembly of the municipal accounts:

1. exports declared without a municipality of origin are redistributed across the declared exporters of the same product and destination in proportion to their volumes, thus conserving national totals (undeclared imports are negligible);
2. the crushing-capacity weight is constructed by dividing each state’s reported capacity total equally among its active plants, as per-plant capacities are not published; the cost of this assumption is quantified in the Technical validation (Section 5.2);
3. the biodiesel-capacity weight attributes each plant’s capacity to soy in proportion to its macro-region’s soy feedstock share;
4. municipal oil and cake output follow from the allocated crush: each year’s oil and cake yields are the ratios of the national oil and cake production to the crushed bean quantity in the FAO balance of that year, giving approximately 20% oil and 75% cake with a 5% processing loss, in line with the yields reported in the literature [13, 14];
5. all geometries are projected to SIRGAS 2000 / Brazil Polyconic (EPSG:5880); each municipality is represented by its administrative centre, and the straight-line distances between these centres provide the transport costs of Appendix D.

The treatment of stock changes, summarized in the last row of Table 2, is derived in Appendix C.

B Livestock systems and feed demand

B.1 Production-system disaggregation

IBGE municipal herd counts are divided into extensive and intensive chickens; extensive, intensive and industrial pigs; and grassland- and mixed-based cattle and buffalo, each split into dairy and meat. The municipal share of each system is the area-weighted zonal sum of the 2010 FAO gridded livestock rasters (GLW3) and the GLEAM ruminant production-system map, normalized across the systems of each species; where a municipality reports a herd in the run year but had no gridded animals in 2010, the state-mean share is substituted. Pigs follow the raster shares directly. For poultry, the extensive share of the total flock forms backyard birds, and the intensive share splits into layers (the reported laying hens) and broilers (the rest; all birds where layers are unreported). Feedlot cattle, reported only in the agricultural census, are the 2006 census counts scaled by the national confinement growth factor of the run year (e.g. $4.38/3.46 \approx 1.27$ for 2013, from ABIEC/Athenagro), holding the municipal pattern fixed; the remaining cattle are split into grassland and mixed systems by the raster shares, with dairy identified by milked cows and meat as the residual. Buffalo use the municipal milked-cow ratio of cattle as the dairy proxy. Each species’ subsystems partition its reported herd exactly.

B.2 Feed demand and national rescaling

Each herd is multiplied by its GLEAM per-animal intake (annual dry-matter intake times the share sourced from soybeans and soy cake, converted to as-fed tonnes at

88% dry matter). One national factor per product then rescales the bottom-up totals to the FAO national feed figures, preserving the spatial and cross-system distribution. Municipal feed demand for beans and cake is the sum over systems. Holding the 2010 system shares fixed over the study period is a simplification: herd sizes remain annual, misclassification affects only how a herd is split across intake intensities, and the national rescaling bounds the aggregate error, which grows with distance from the 2010 reference year.

C National balancing and trade reconciliation

For each product, the national commodity balance imposes the supply–use identity: production, imports and stock withdrawals (supply) equal exports, food, feed, seed, processing and other uses (use). The trade harmonisation collapses destinations absent from the FABIO/FAOSTAT bilateral list into a rest-of-world aggregate; it aligns classifications only and applies no rescaling. The balance items are rescaled proportionally, with each item’s municipal distribution as weight, after the national residual has been moved into the stock term. A net national stock withdrawal is carried on the supply side: the drawdown is distributed across municipalities by the stock weights (storage capacity for beans, production for oil and cake) and added to municipal supply. A drawdown is a genuine source of current-year supply, and this treatment alone keeps every municipal use non-negative, as required by the re-export inversion of Appendix E. Net additions remain on the use side. As each year is solved independently, tonnes supplied from a drawdown, physically grown in earlier years, inherit the current year’s soy-area intensity (Appendix G). Finally, each municipality’s product balance defines its surplus, $\max(\text{supply}_m^p - \text{use}_m^p, 0)$, and deficit, $\max(\text{use}_m^p - \text{supply}_m^p, 0)$, which drive the inter-municipal flows of Appendix D; the two totals match by construction after the rescaling, as required by the balanced transportation problem. The national residual itself cannot be resolved into flows; it is retained as FABIO’s signed balancing column and re-enters the model as a dedicated final-demand category (Appendices F and G).

D Optimal-transport formulation

D.1 The Euclidean model

For each product p , the inter-municipal flow matrix $\mathbf{F}^p$ solves the balanced transportation problem

$$\mathbf{F}^p = \arg \min_{\mathbf{F} \geq 0} \sum_{m,n} F_{mn} d_{mn} \quad \text{s.t.} \quad \sum_n F_{mn} = \text{surplus}_m^p, \quad \sum_m F_{mn} = \text{deficit}_n^p, \quad (\text{D.1})$$

where d_{mn} is the straight-line distance between the administrative centres of municipalities m and n (Appendix A) (the two margins are balanced beforehand by proportionally scaling the larger one). The problem is solved exactly by the network-simplex algorithm (R package `transport` [15]), separately for beans, oil and cake.

D.2 The multimodal network variant

To test whether the straight-line metric distorts the allocation, the same municipal surpluses and deficits are re-solved on a multimodal representation of Brazil’s logistics network, generalizing the single-year (2013) intermodal model of Trsek [10] to the years 2010–2022. The variant has three building blocks:

1. the network: roads (OpenStreetMap, simplified to a 5 km grid), railways (ANTT lines and stations) and waterways (DNIT rivers with ANTAQ ports). A tonne may travel by road directly, or by road to a rail or waterway terminal, along that network, and onwards by road to its destination;

2. the costs: every leg is priced at a unit cost per tonne-km that depends on the mode (road 0.013–0.174, rail 0.006–0.065, water 0.004–0.032 USD), and changing modes carries a switching charge (0.74–2.57 USD per tonne); the central runs use the midpoints of these ranges, taken from the freight-cost literature compiled in [10];
3. the capacities: a rail or waterway corridor cannot carry more of a product than it was observed to carry, from ANTT origin–destination rail records (all study years) and ANTAQ port statistics (observed from 2017 onwards; earlier years are proxied with the 2017 records, as ANTAQ’s pre-2017 archives are currently unavailable).

The resulting minimum-cost flow problem is solved with the open-source HiGHS solver; the 2021 problem is infeasible on the current inputs and is excluded from the comparison.

E Origin tracing and re-export disentangling

E.1 Linking origin municipalities to importing countries

The transport flows of Appendix D record only what moves between municipalities, so the part of each municipality’s supply that stays and is used at home is added on the diagonal,

$$\tilde{F}_{mn}^p = F_{mn}^p + \mathbf{1}[m = n] (\text{supply}_m^p - \text{surplus}_m^p), \quad (\text{E.1})$$

so that the rows of $\tilde{\mathbf{F}}^p$ sum to total municipal supply and its columns to total municipal use. Every exported tonne is then unwound to its origin (Figure 3),

$$X_{mj}^p = \delta_m^p \sum_n \phi_{mn}^p E_{nj}^p, \quad (\text{E.2})$$

where E_{nj}^p are the observed exports of municipality n to country j ; the sourcing share $\phi_{mn}^p = \tilde{F}_{mn}^p / \sum_{m'} \tilde{F}_{m'n}^p$ is the fraction of n ’s total use that originated in municipality m (including $m = n$); and the domestic-origin share $\delta_m^p = (\text{supply}_m^p - \text{imports}_m^p) / \text{supply}_m^p$ strips out quantities of foreign origin (withdrawal-supplied tonnes count as domestic under the supply-side stock treatment; Appendix C). Local consumption is traced with the same formula through a synthetic domestic column added to the export table (for beans net of the crush, which is traced through the oil and cake chains instead), and oil and cake are traced one step further back, to the bean-growing municipalities, by premultiplying their origin-to-destination matrices with the bean sourcing shares and the bean domestic-origin share.

E.2 Re-export disentangling

To attribute internationally traded soy to its true origin, bilateral trade is disentangled with the standard Leontief-type inversion of physical trade-linkage accounting [11]: trade is normalized by destination use, the infinite re-export chain is summed by matrix inversion, and retaining the domestically produced quantities yields the origin-to-consumption matrix,

$$\mathbf{R} = \mathbf{T} \hat{\mathbf{u}}^{-1}, \quad (\mathbf{I} - \mathbf{R})^{-1} = \sum_{k=0}^{\infty} \mathbf{R}^k, \quad (\text{E.3})$$

where $\mathbf{T}$ is the bilateral trade matrix, $\hat{\mathbf{u}}$ the diagonalized vector of total use by region, R_{ij} the share of region j ’s use supplied by imports from region i , and $\mathbf{R}^k$ carries the flows re-exported k times, so the inverse sums chains of every length. The departure from the published method is the regional dimension: for the soy products Brazil is replaced by its 5,570 municipalities, enforced throughout the processing of the trade data so that no national Brazilian record re-enters the inversion. The inversion treats the soy available in a region as one well-mixed pool: local production and incoming shipments blend in proportion to their volumes, and every user of the pool receives the

same origin mix, as is natural for a bulk commodity blended in silos. Which municipality exports to which country is observed in the customs data, so the assumption does not touch that step; it fills in only where an exporting municipality’s soy came from within its catchment, and how soy is passed on when a country re-exports it. What it cannot represent is a buyer who deliberately selects soy of a specific origin, as European importers do under the Amazon Soy Moratorium: such origin-specific contracting is not resolved. The inversion is computed exactly by a sparse solver, with a dense fallback and, as a last resort, a small diagonal ridge to lift a genuine unit eigenvalue (a commodity whose recorded global trade is entirely re-exported); the years requiring the ridge are documented with the released code.

F FABIO backends and temporal coverage

The FABIO v2 build exists only from 2010 onwards; for 2001–2009 the pipeline switches to four inputs from the FABIO v1.1 build: the full bilateral trade detail, the land-use extension, the feedstock optimisation results, and the v1 balance sheets (the v2 balance file contains only placeholder rows before 2010). Version bridges map through stable FAO item codes, and the v1.1 land-use extension, which reports total agricultural land rather than v2’s cropland, is converted to the per-process intensity vector through the same mapping; soy is cropland under both definitions, so the soy land footprint is unaffected, but non-soy land intensities are not comparable across vintages. The seam is quantified on the overlap years 2010–2013, run through both chains end to end: in the two clean overlap years totals differ by -6% (2011) and -8% (2013, v1.1 low) with cross-country correlations of consumer footprints of 0.988 and 0.992 (the 2012 v1.1 build halts at the re-export conservation check, and 2010 diverges through a disagreement between the balance-sheet vintages), so pre-2010 levels are reported unadjusted. One usage caveat: Brazil’s domestic footprint is about a third lower on v1.1, so cross-vintage statements about Brazil-domestic use are restricted to 2010 onwards.

G Land-use footprint accounting

G.1 Footprint calculation

With $e_i = \text{land}_i/x_i$ the land intensity of process i (its direct soy area per unit of output; Section 3.6), the footprint is the standard intensity–Leontief–demand product

$$\mathbf{H} = \hat{\mathbf{e}}\mathbf{L}\mathbf{Y}, \quad (\text{G.1})$$

where $\hat{\mathbf{e}}$ is the diagonal matrix of the land intensities, $\mathbf{L}$ the Leontief inverse of the nested system, $\mathbf{Y}$ final demand aggregated to consumer countries, and H_{ij} the hectares of soy land in origin municipality i embodied in country j ’s consumption. Stock additions and the balancing category are removed from final demand before aggregation: stock drawn down in year t embodies land use of earlier years, and retaining it as negative year- t demand would attribute negative footprints to producer regions in drawdown years; EXIOBASE inventory changes are treated analogously. The food system and the non-food background traced through EXIOBASE [7] contribute additively to each consumer’s total. Footprints by consumed product are obtained by restricting $\mathbf{Y}$ to selected commodity groups (e.g. dairy, meat) before applying (G.1).

G.2 Origin attribution and grid refinement

The biome assignment intersects each municipality’s interior point with the biome layer; points falling outside every biome polygon are assigned the nearest biome. A destination’s frontier share is the fraction of its origin footprint located in frontier municipalities, with the frontier as defined in Section 3.6. For visualization, per-municipality embodied production can optionally be distributed onto 30 m MapBiomass

soy pixels, each pixel of a municipality carrying that municipality’s per-pixel share for the given consumer country; this refinement is illustrative and does not affect the reported results.

H Validation and uncertainty analysis

H.1 Benchmark and baseline

The comparison sample is built from the origin-to-importer flows X_{mj}^p of Appendix E, the Trase composite, and the downscaling baseline, defined as national exports to each destination allocated to municipalities in proportion to their share of domestic-origin soybean supply: every importer draws from the same municipal map, importers differ only in total volume, and the baseline contains no supply-chain structure by construction. All three sources are converted to bean equivalents with the national crush ratio and summed over beans, oil and cake. Trase flows with unknown origin municipality are excluded, destinations are harmonized to the bilateral-trade classification, and the sample comprises all soy-producing municipalities crossed with the destinations recording positive exports in both benchmark and model. Brazil, which enters the tables as the destination of domestic use, is excluded wherever importers are compared or weighted.

H.2 Agreement by importer

Figures 12 and 13 unpack the volume-weighted average reported in the Technical validation section at the two ends of the volume distribution: the four importers that together take 70% of the Trase-recorded 2010–2022 volume (China, the Netherlands, Thailand and Spain), and the five smallest importers with matched flows in every year (Cuba, Israel, the United Arab Emirates, Greece and Australia, 0.1–1.5 Mt each over the whole period). Each figure shows the mean correlation over its importer set and, below it, the full per-importer scores per year for the three allocation variants.

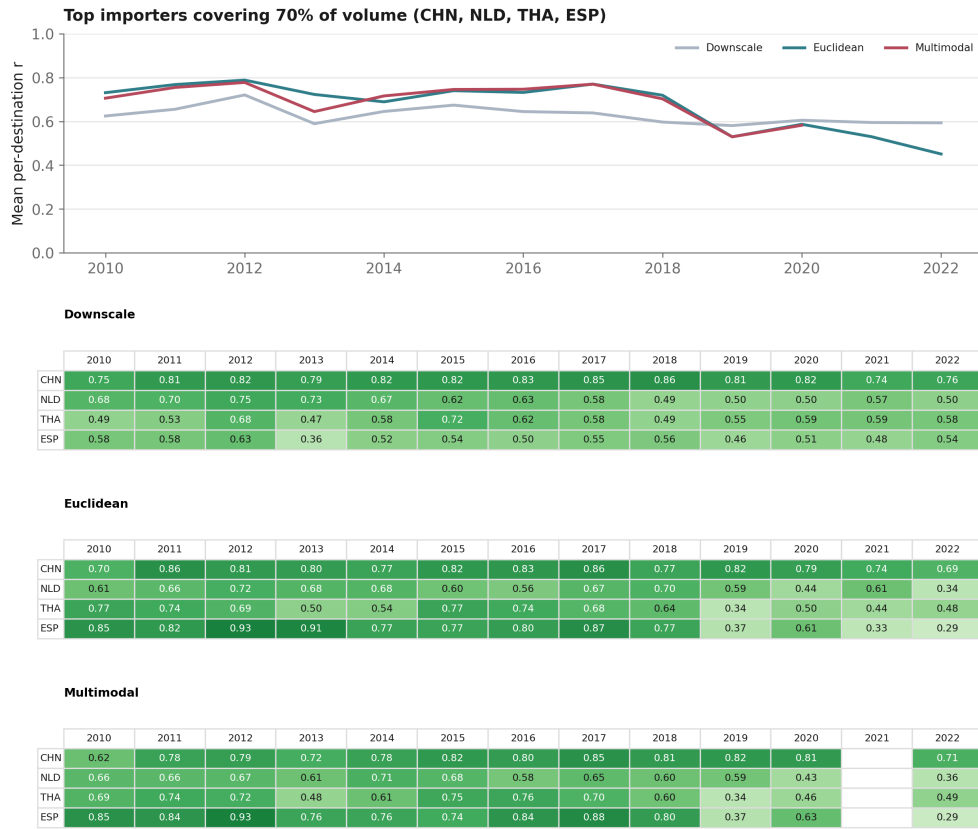


Fig. 12 Per-destination correlation with Trase for the four importers covering 70% of the 2010–2022 volume. The line plot shows the mean over the four importers; the tables give the per-importer scores per year for each allocation variant (colour scale from 0, white, to 1, dark green; the multimodal variant has no 2021 solve).



Fig. 13 As Figure 12, for the five smallest importers matched in every year. At this volume level, three orders of magnitude below China, the correlations scatter from slightly negative to above 0.9 with no persistence from year to year: individual small-importer results carry little information.

H.3 Perturbation recipes of the input-sensitivity screen

Crush location. Each state's fixed capacity total is redistributed over its plant municipalities in proportion to a weight k_m : one in the baseline (equal split), municipal soybean production in the production-proportional variant, and $k_m \sim \text{Exp}(1)$ drawn independently per plant in the random variants, which makes every split of the state total equally likely (a uniform Dirichlet distribution over the shares); ten seeds give ten capacity maps. The allocation-sharpness variants leave the split untouched and raise the allocation weight to cap_m^γ with $\gamma \in \{0.5, 2\}$.

Feed rates. The per-head bean and cake rates of every livestock system are multiplied by independent mean-one lognormal factors,

$$f = \exp(\mathcal{N}(-\sigma^2/2, \sigma^2)), \quad \sigma = 0.3, \quad (\text{H.1})$$

drawn separately for the bean and the cake rate of each system; ten seeds give ten parametrisations. Because national feed totals are rescaled to the FAO commodity balance after the multiplication (Appendix B), a factor common to all systems cancels exactly, and only the relative system mix moves the spatial allocation. The herd-composition variant replaces the 2010-frozen municipal system shares with state averages.

I Data access links

Dataset names below match Table 1, which gives provider, role and coverage; this annex adds only the access links. All URLs were last verified in July 2026. Brazilian government portals may reject automated requests but resolve in a browser. A programmatic downloader covering the machine-retrievable sources ships with the code (`code/download_data.py`).

Brazilian subnational statistics

- **Soy production, area** (SIDRA table 1612): <https://sidra.ibge.gov.br/tabela/1612>; API pattern <https://apisidra.ibge.gov.br/values/t/1612/n6/all/v/109,216,214/p/{year}/c81/2713>.
- **Population** (tables 6579/9514), **herds** (table 3939), **milk cows** (table 94), **feedlot cattle** (tables 919/6911): same API pattern with the respective table number.
- **Grain storage** (table 278): <https://sidra.ibge.gov.br/tabela/278>.
- **Feedlot confinement series** (ABIEC Beef Report; manual download): <https://www.abiec.com.br/publicacoes/>.
- **Crush/refining capacity** (manual download): <https://abiove.org.br/estatisticas/>.
- **Biodiesel capacity** (Statistical Yearbook table 2.6; manual download): <https://www.gov.br/anp/pt-br/centrais-de-conteudo/publicacoes/anuario-estatistico/>.
- **Soy-oil acquisition (POF)**: <https://www.ibge.gov.br/estatisticas/sociais/educacao/9050-pesquisa-de-orcamentos-familiares.html>.
- **Municipal trade**: per-year files <https://balanca.economia.gov.br/balanca/bd/comexstat-bd/mun/> (EXP-`{year}`_MUN.csv, IMP-`{year}`_MUN.csv); lookup tables <https://balanca.economia.gov.br/balanca/bd/tabelas/> (PAIS.csv, UF_MUN.csv). Directory listings are blocked; the files themselves download normally.

National balances and feed parameters

- **Commodity balance (FBS)**: <https://www.fao.org/faostat/en/#data/FBS>; bulk file [https://bulks-faostat.fao.org/production/FoodBalanceSheets_E_All_Data_\(Normalized\).zip](https://bulks-faostat.fao.org/production/FoodBalanceSheets_E_All_Data_(Normalized).zip).
- **Detailed trade matrix**: bulk file [https://bulks-faostat.fao.org/production/Trade-DetailedTradeMatrix_E_All_Data_\(Normalized\).zip](https://bulks-faostat.fao.org/production/Trade-DetailedTradeMatrix_E_All_Data_(Normalized).zip).
- **Gridded livestock (GLW3)**: <https://dataverse.harvard.edu/dataverse/glw>.
- **Production systems, feed (GLEAM)**: <https://www.fao.org/gleam/resources/en/>.

Transport network layers

- **Road network**: <https://download.geofabrik.de/south-america/brazil.html>.
- **Waterways**: <https://www.gov.br/dnit/pt-br/assuntos/atlas-e-mapas>; download tool VGeo <https://servicos.dnit.gov.br/vgeo/> (may reject automated requests).
- **Rail lines, stations, cargo**: <https://dados.antt.gov.br/dataset/transporte-ferroviario-de-cargas-e-passageiros>.
- **Ports, port cargo**: <https://web.antaq.gov.br/anuario/> (temporarily unreachable at the July 2026 check; re-verify before use).

Geographies

- **Municipal boundaries** (geobr): <https://github.com/ipeaGIT/geobr>; <https://cran.r-project.org/package=geobr>; localities API <https://servicodados.ibge.gov.br/api/v1/localidades/municipios>.
- **State boundaries**: <https://gadm.org/>.
- **Biomes 1:250k**: <https://www.ibge.gov.br/geociencias/informacoes-ambientais/estudos-ambientais/15842-biomas.html>.

- **Soy land tiles** (via Google Earth Engine): <https://mapbiomas.org/>; export toolkit <https://github.com/mapbiomas-brazil/user-toolkit>.

Global models and reference series

- **FABIO v2 / v1.1**: <https://github.com/fineprint-global/fabio>; v1.1 archive DOI 10.5281/zenodo.2577066.
- **EXIOBASE 3** **pxp**: <https://www.exiobase.eu/>; data DOI 10.5281/zenodo.3583070.
- **Trase composite v2.6.1**: <https://trase.earth/>; supply-chain dataset DOI 10.48650/DCE3-JJ97 (v2.6).

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